Parth Rathod

EV Systems Engineer

RESUME_1259

Location: Vadodara, Gujarat, India | Email: parth.rathod.resume_1259@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 3 years

PROFESSIONAL SUMMARY

EV Systems Engineer with 3 years of experience in ev systems, using Battery Management Systems, CAN Bus, MATLAB, Simulink. Experienced in model-based design, automotive safety and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Python, HIL, CAN Bus, EV Powertrain, Motor Control, AUTOSAR, Battery Management Systems, SOC Estimation, ISO 26262, Simulink, MATLAB, Diagnostics

Methods: Model-based design, Automotive safety, Vehicle networks

PROFESSIONAL EXPERIENCE

EV Systems Engineer | JuniperWorks Systems | Jul 2023 - Present

- Improved SOC estimation error by 26% after parameter calibration
- Validated 631 HIL fault-injection scenarios for protective responses
- Lowered peak powertrain losses by 29% in control-model experiments
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

PROJECTS

Battery Pack Health Monitor | Simulink, Python, CAN Bus, AUTOSAR

- Implemented SOC estimation checks and temperature fault diagnostics.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

Electric Drive Controller | Diagnostics, Simulink, Python, Battery Management Systems

- Modeled torque request logic and motor protection interlocks.
- Documented acceptance criteria and evaluated the design using automotive safety and vehicle networks.

EDUCATION

B.Tech in Electrical Engineering, Aarohan Institute of Engineering, 2023
Relevant Methods: Model-based design, Automotive safety, Vehicle networks

CERTIFICATIONS

EV Powertrain Fundamentals (2023)